DEL MAR
EST. SEAFOODS 1988

PROJECT BRIEF & TECHNICAL REQUIREMENTS · 项目简报与技术要求

VERTICAL PLATE FREEZER PROJECT

立式平板冻结项目 — 技术要求简报

Prepared by Del Mar Seafoods for Jiangsu Wolfking Tech Co., Ltd. and APCCO. English edition; a Chinese edition of the same document is issued separately.

DOCUMENT	DATE	SITE	STATUS
DMS-VPF-001 · Rev A	30 October 2026	Watsonville, California, USA	Issued for design start

1 PURPOSE

This document brings together everything Del Mar Seafoods has decided for its vertical plate freezer project in Watsonville, California, and sets out how Del Mar, Wolfking and APCCO divide the work. It is the starting point for Wolfking's design. It replaces nothing already agreed by email; it collects it in one place.

Each requirement carries a status so there is no doubt about who owns it:

CONFIRMED decided by Del Mar; fixed unless Del Mar changes it

WOLFKING Wolfking to design or propose

APCCO APCCO owns or will confirm

OPEN not yet decided; owner and timing listed in §12

HOW WE WORK

One shared email thread ("Del Mar plate freezing projects", started by Joe on 30 October 2026) for all project communication, with Del Mar, Wolfking and APCCO on every message. APCCO's engineers will ask many detailed questions. These are for the California permit, not a sign of problems; please answer them directly.

2 PARTIES AND ROLES

PARTY	ROLE IN THIS PROJECT
Del Mar Seafoods, Inc. Watsonville, CA — owner and operator	Buys the equipment, owns the site and the process. Joe Cappuccio (President): commercial decisions, product and schedule. Kevin Cappuccio: project engineering, electrical, technical liaison. Del Mar approves the block size, the layout and the final design before manufacturing.
Jiangsu Wolfking Tech Co., Ltd. Nantong, Jiangsu — designer and manufacturer	Designs and manufactures four vertical plate freezers with hydraulic power unit(s) and controls, and the complete ammonia refrigeration package for them (packaged engine room preferred, see §6). Supplies design drawings, calculations, certificates and documentation needed for the California permits. Sends two technicians to site to guide APCCO's crew during installation and to lead commissioning. Supports Del Mar's phase-2 automation (loading, unloading, bagging and boxing) with its partners.
APCCO — Applied Process Cooling Corp. Fresno, CA — licensed refrigeration contractor and engineer	Reviews Wolfking's design for California compliance, prepares and stamps permit drawings and calculations, files the permits with the City of Watsonville and other agencies. Installs all delivered equipment: rigging and setting, ammonia piping and insulation, electrical connections, controls integration. Performs start-up with Wolfking and provides ongoing maintenance and service. APCCO does not manufacture freezers; it is Del Mar's local partner for installation and compliance and will work directly with Wolfking's engineers.

CERTIFICATES, AGREED UP FRONT

To avoid the problem Wolfking described from a previous export (documents demanded at the port after shipment), APCCO will give Wolfking a written list of every certificate, test report and document required for this installation before manufacturing starts, and Wolfking will confirm which it can provide. APCCO

3 PRODUCTS AND PRODUCTION TARGETS

TWO PRODUCTS ON THE SAME FREEZERS CONFIRMED

- **Whole California market squid** (*Doryteuthis opalescens*), plate-frozen in blocks. This is Del Mar's test product for the China market: if Chinese buyers accept plate-frozen squid, Del Mar will convert more production to this method. The Monterey Bay squid season starts in May, so the line must be running by May 2027.
- **Pet-food raw material**: ground squid, a paste-like mince, frozen in the same blocks. This runs year-round under an existing contract and is the base load that pays for the equipment.

DAILY PRODUCTION CONFIRMED

45 t	2 × 10 h	4	8	140
per day (50 short tons), pet-food contract	shifts per day + 4 h sanitation	vertical plate freezers, identical	batches per freezer per day (4 per shift)	blocks of 10 kg per batch (target)

Del Mar's cycle assumption: 90 minutes freezing plus 30 minutes to unload and reload, so one batch every 2 hours and four batches per shift. 45 t ÷ 8 batches ÷ 4 freezers gives about 1.4 t, or 140 blocks of 10 kg, per freezer per batch. 140 blocks is the goal; fewer is acceptable if the design cannot reach it, but please try.

BACKGROUND AND WHAT COMES NEXT

Today Del Mar freezes large volumes of squid in cartons on racks in blast freezers. That is very labor-intensive at California labor rates. The goal of this project is block freezing with far less handling. The four freezers are a pilot: if the product and the economics work, Del Mar will add freezers in multiples of this design, so Wolfking should design for repeatability. Phase 2 (planned for 2027–2028) is automation: automatic loading of the freezers, automatic unloading, bagging and carton packing with a check-weigher. The phase-1 freezers must be ready to accept that automation later (see §5).

4 BLOCK SPECIFICATION

ITEM	REQUIREMENT	STATUS
Block mass	10 kg nominal. Squid is about 20% less dense than water, so a 12-litre block gives roughly 10 kg (10 kg of water = 10 L).	CONFIRMED
Block volume	12,000,000 mm^3 (12.0 L) nominal.	CONFIRMED
Thickness	75 mm. Fixed; this sets the freezing time.	CONFIRMED
Face size	Target about 535 × 300 mm (535 × 300 × 75 = 12.04 L). Del Mar is flexible on the exact face dimensions, but whatever is chosen becomes the Del Mar standard block for all future freezers. Wolfking to propose the face size that best fits its plate extrusions and still gives 12 L; Del Mar approves before plates are ordered.	WOLFKING
Blocks per station	Working assumption from the Qingdao meeting: two divider plates per station, three blocks side by side. Wolfking to confirm against the chosen face size, plate height and block orientation (which dimension is vertical).	WOLFKING
Blocks per batch	140 per freezer is the target (117–120 was discussed at the show; Joe's email of 30 October raises it to 140). Design as close to 140 as practical and state the resulting plate count and frame length.	WOLFKING
Freezing time	Del Mar assumes 90 minutes for a 75 mm block. Wolfking to confirm the time to −18 °C core at the design suction temperature, for whole squid and for ground squid paste, and to state the assumptions (product in-feed temperature, plate temperature, contact).	WOLFKING
Drainage of free liquid	Whole squid releases a lot of water and ink when loaded and in the first minutes of freezing. The bottom of each station must let this free liquid drain away rather than trap it, so the customer receives a block of squid, not a block of ice. But the pet-food paste must not run out of the station. Wolfking to propose a bottom design that handles both (for example a drain gap or holes sized to pass thin liquid but hold paste, or an adjustable arrangement). Del Mar will send product photos or samples if useful.	WOLFKING

5 VERTICAL PLATE FREEZER REQUIREMENTS

ITEM	REQUIREMENT	STATUS
Quantity and type	Four identical vertical plate freezers, top-loaded, installed indoors in the processing room. Hydraulic plate closing/opening, hot-gas release and hydraulic block lift. Block discharge to one side of the freezer onto a table or conveyor; the discharge side is fixed in the layout (§8).	CONFIRMED
Capacity	Per §3 and §4: 75 mm blocks, 3 per station, 140 per batch target, 8 batches per day.	CONFIRMED
Loading, phase 1	Manual or semi-automatic. Del Mar is considering a top conveyor with gates for whole squid and hose/pump filling for paste; Wolfking's advice on the simplest reliable method for 140 stations is welcome. Final choice is Del Mar's.	OPEN
Ready for phase-2 automation	The freezers must accept Wolfking's automatic loading and unloading system later without being replaced. Design in the interface now: top clearance, mounting points or rails for a loading gantry, the block discharge conveyor interface, and spare control I/O. Wolfking to state what provisions it needs built in.	WOLFKING
Unloading and packing	After the lift, blocks go to bagging and carton packing (block in a bag, bag in a box), with a check-weigher planned for phase 2. Wolfking offered a partner's bagging and case-sealing machine; please include it as a separately priced option.	WOLFKING
Materials and hygiene	Plates: aluminium alloy (extruded), food-contact grade. Product zone and splash zone, including everything above the plates: stainless steel or aluminium, cleanable, no galvanized surfaces. Structural base below the product zone: Wolfking to propose stainless steel or hot-dip galvanized with the price difference; the plant handles salt water and chlorinated sanitizers daily.	WOLFKING
Hydraulic power unit	Motor 460 V / 60 Hz (§7). Located outside the cold and wet zone. Oil and seals compatible with NSF H1 food-grade hydraulic oil, and rated for −35 °C at the cylinders and hoses. Wolfking to state motor kW, working pressure, and whether it proposes one unit per freezer or one shared unit.	WOLFKING
Refrigerant connections	Flexible connections to the moving plates: stainless steel, rated for ammonia, replaceable from outside. No copper or brass anywhere in the ammonia circuit.	CONFIRMED

ITEM	REQUIREMENT	STATUS
Safety	Emergency stop, guarding of the pinch points at the open top, lockable electrical disconnect, means to isolate and bleed hydraulic energy. Guarding to US practice; APCCO will review (§9).	APCCO
Documentation	In English: general arrangement drawings, P&ID, electrical schematics, operation and maintenance manual, spare-parts list with a recommended two-year spares package (hoses, seals, valves). 3D models (STEP) of the freezer and skids for Del Mar's layout work.	CONFIRMED

6 REFRIGERATION PACKAGE

Wolfking designs and supplies the complete refrigeration system for the four freezers. APCCO reviews it for California compliance, installs it and maintains it.

ITEM	REQUIREMENT	STATUS
Refrigerant	Anhydrous ammonia, R717. California no longer allows high-GWP HFC refrigerants in new systems of this size, so ammonia is the practical choice.	CONFIRMED
Scope	Compressor package, condenser, vessels, ammonia pumps if a recirculated design, valve stations, safety reliefs, controls and control panel, and the interconnecting piping design. APCCO installs the field piping, insulation and electrical.	CONFIRMED
Arrangement	Del Mar's preference, discussed in Qingdao: a packaged or containerized engine room located outdoors, next to the building, with the freezers indoors and insulated piping between them. This saves building space and keeps compressor noise outside. Del Mar and APCCO meet at the Watsonville site in early November to settle the location and the piping route; Wolfking to send its standard packaged engine room dimensions and a concept layout before then. Alternative Joe asked about: a small self-contained refrigeration unit on each freezer; Wolfking to comment on feasibility, charge per unit and cost compared with one central package.	OPEN

ITEM	REQUIREMENT	STATUS
Ammonia charge	Minimum charge is a design goal. California's CalARP program applies at 500 lb (227 kg) of ammonia on site, and staying below that line simplifies permitting considerably. Wolfking to calculate the total charge for its proposed design. Del Mar understands it may not be possible at this capacity; if not, state the charge so APCCO can plan the filing.	WOLFKING
Design conditions	APCCO's early estimate, made before the batch size was raised to 140 blocks: about 45 TR (158 kW) at −30 °F (−34 °C) suction for four freezers, pumped recirculation assumed at 4:1. Wolfking to provide its own load calculation for the final batch size, including the peak pull-down load with staggered batches, and the resulting compressor and condenser selection.	WOLFKING
System type	Wolfking to state: pumped recirculation or other feed method and the recirculation ratio; suction and condensing temperatures; condenser type (evaporative or air-cooled; water use and noise both matter at this site); receiver or critical-charge design; defrost method (hot gas assumed) and the hot-gas requirements of the freezers.	WOLFKING
Valves and controls	Wolfking to provide a valve and device schedule with brands, and to state which valves ship mounted on the freezers, which ship loose, and which it expects APCCO to supply. State whether the freezer control panel controls the refrigeration valves. APCCO will confirm which brands are acceptable for service in California.	WOLFKING
Pressure design	Pressure vessels ASME Section VIII Division 1 with U-stamp and National Board registration. Piping to ASME B31.5. Design pressures per IIAR 2; the plates and their piping see discharge-side pressure during hot-gas defrost, so they must be rated for it (APCCO will confirm the required values, typically 250 psig / 17.2 bar). Vessel welds radiographed where the California vessel code requires it. Material certificates and welder qualifications available on request.	APCCO

7 ELECTRICAL

ITEM	REQUIREMENT	STATUS
Site power	480 V, 3-phase, 60 Hz (480Y/277 V, grounded wye). All motors nameplated 460 V / 60 Hz. There is no 380 V and no 220 V single-phase supply in the United States.	CONFIRMED

ITEM	REQUIREMENT	STATUS
Control power	120 VAC, 60 Hz, single-phase, from a control transformer inside each panel; 24 VDC for instruments, I/O and solenoid valves. No 220 V single-phase components (heaters, fans, lights, coils).	CONFIRMED
Control panels	Built and labelled to UL 508A (a UL or ETL listing to UL 508A), with the short-circuit current rating marked. NEMA 4X stainless enclosures in washdown areas. Components UL-listed or UL-recognized (Schneider, ABB, Siemens, Eaton, Allen-Bradley or equivalent). Wiring practice and colour code to NFPA 79; labels in English.	CONFIRMED
Motors	NEMA Premium / IE3 efficiency, which US Department of Energy rules require for imported motors even when they are built into a machine. Hydraulic pump and fan motors sized for 60 Hz speed, not relabelled 50 Hz units.	CONFIRMED
Load list, needed first	Per freezer and per skid: kW, full-load amps at 480 V, starting method and inrush, and the proposed number and size of feeders. Del Mar needs this to apply to the utility (PG&E) for service, which can take longer than manufacturing.	WOLFKING
Controls	PLC and HMI with English text. Remote access for Wolfking support is welcome; Del Mar provides the internet connection. Wolfking to propose the PLC brand; Siemens or Allen-Bradley preferred for service in the US.	WOLFKING

8 SITE: WATSONVILLE, CALIFORNIA

- **Location.** Del Mar Seafoods, Watsonville, California, on Monterey Bay about 150 km south of San Francisco. Mild coastal climate. APCCO will supply the design ambient temperatures for the condenser (summer design conditions are roughly 29 °C dry bulb / 18 °C wet bulb; to be confirmed). APCCO
- **Layout.** Freezers indoors in a processing room; packaged engine room outdoors adjacent to the building; piping route and lengths set at the Del Mar–APCCO site meeting in early November. Wolfking to state its preferred maximum distance between engine room and freezers. OPEN
- **Seismic.** Watsonville is in a high-seismic zone (the 1989 Loma Prieta earthquake was centred nearby). Every piece of equipment needs anchorage designed for California seismic loads. Wolfking to provide weights, centres of gravity and anchor-bolt locations for the freezers, skids and vessels; APCCO's engineer does the anchorage calculations. WOLFKING
- **Floor and foundations.** Wolfking to provide footprints and point loads for the freezers and the engine room package. WOLFKING

- **Utilities available.** 480 V 3-phase power, potable water, drains, internet. Compressed air to be confirmed.
- **Transport.** Ocean freight to the Port of Oakland or Long Beach, then road to Watsonville. A packaged engine room should fit standard container or legal road dimensions; APCCO will advise on rigging access at the site. Shipping terms to be agreed in the proposal. OPEN

9 REGULATORY NOTES FOR THE DESIGN PHASE

Only the points that change what Wolfking designs. APCCO is the authority on all of them and will give the specific values; these notes are so nothing is a surprise.

1. **Permits.** The City of Watsonville issues building, mechanical and electrical permits on drawings and calculations stamped by a California-licensed professional engineer (APCCO's P.E.). Wolfking supplies the design; APCCO adapts, stamps and files. Permits can take longer than manufacturing, which is why drawings come first.
2. **Ammonia system design.** ANSI/IIAR 2-2021 (design of closed-circuit ammonia systems) with IIAR 4, 5 and 6 for installation, start-up and inspection, plus the refrigeration chapter of the California Mechanical Code. These set design pressures, relief valves, valve arrangement and labelling, emergency shutdown, and ammonia detection and ventilation for any indoor machinery room. An outdoor packaged engine room reduces the indoor requirements.
3. **Ammonia inventory thresholds.** California's CalARP program applies from 500 lb (227 kg); federal OSHA PSM and EPA RMP from 10,000 lb. Design for minimum charge and document it.
4. **Vessels and piping.** ASME Section VIII Div. 1 U-stamp and National Board registration for vessels; ASME B31.5 for refrigeration piping; Cal/OSHA pressure-vessel permits, with radiography of vessel welds where required. Material certificates (MTRs) and welder qualifications to ASME IX must be available.
5. **Electrical.** California Electrical Code (NEC). Every piece of electrical equipment must be listed by a US testing laboratory (UL, ETL, CSA): panels to UL 508A, motors and components listed or recognized, motors to DOE efficiency levels. Unlisted equipment needs a field evaluation, which is slow and expensive, so design it out.
6. **Refrigerant.** California (CARB) prohibits high-GWP HFCs in new refrigeration systems with more than 50 lb charge; ammonia is the compliant choice and is already decided.
7. **Seismic.** California Building Code and ASCE 7: equipment anchorage and restraint designed for the site. Skids, vessels and freezer frames need designed anchor points.
8. **Food safety.** FDA FSMA: seafood HACCP (21 CFR 123) and animal-food preventive controls (21 CFR 507). Product-contact surfaces must be corrosion-resistant and cleanable; lubricants that could reach product must be NSF H1; no flaking coatings above product.
9. **Worker safety.** Cal/OSHA: machine guarding, emergency stops, lockout/tagout provisions, English labels and warnings.
10. **Noise.** Outdoor equipment next to the building must meet the city's noise limits at the property line. This affects condenser fan selection and whether the package needs acoustic treatment. Del Mar and APCCO will give the limit once the location is fixed.

10 SCHEDULE

WHEN	WHAT	WHO
Week of 2 Nov 2026	This brief issued. Del Mar sends product photos and any samples. Del Mar and APCCO meet at the Watsonville site on the engine room location and layout. Wolfking starts design.	All
As soon as possible	Preliminary general arrangement, P&ID, electrical load list, ammonia charge estimate and proposal (§11). Del Mar and APCCO review; permit work starts on the preliminary design.	Wolfking
After design approval	Manufacturing of freezers and refrigeration package. Wolfking's estimate at the show: about 100 days.	Wolfking
Shipping and installation	Ocean transit, customs, rigging, installation, commissioning. APCCO to confirm installation duration and crew.	APCCO + Wolfking technicians
May 2027	Target: installed and commissioned for the start of the squid season. June 2027 is the latest acceptable date. Joe has told the pet-food customer six months in total.	All
2027–2028	Phase 2: automatic loading and unloading, bagging and case packing, check-weighing.	Wolfking + partners

11 WHAT WE NEED FROM WOLFKING FIRST

1. **Preliminary general arrangement of one freezer:** dimensions, plate count, station layout, proposed block face size, loading height, discharge side, weights.
2. **Freezing-time calculation** for 75 mm blocks of whole squid and of squid paste, with assumptions.
3. **Refrigeration concept:** system type, P&ID, equipment list, ammonia charge estimate, packaged engine room concept with dimensions and weight.
4. **Electrical load list** and proposed one-line diagram; control-panel specification and the listing approach (UL or ETL).
5. **Vessel list** with ASME and National Board status; welding and QA documentation plan; list of certificates Wolfking can supply (to be matched against APCCO's list, §2).
6. **Commercial proposal:** scope split between Wolfking supply and APCCO installation, price, delivery, payment terms, warranty, recommended spares package, and terms for the on-site technicians (number of people, days, travel).
7. **3D models** (STEP) of the freezer and engine room package when available, for Del Mar's layout.
8. **Reference list** of US installations, including the Oregon and Los Angeles customers discussed in Qingdao.

9. **Phase-2 budget concept** for automatic loading and unloading and for the partner bagging and case-packing equipment, so the interfaces are designed into the freezers now.

12 OPEN ITEMS

ITEM	OWNER	NEEDED BY
Final block face dimensions (around 535 × 300 × 75 mm, 12 L)	Wolfking proposes, Del Mar approves	Before plates are ordered
Blocks per batch (140 target) and plate count	Wolfking	With preliminary GA
Freezing time confirmation	Wolfking	With preliminary GA
Station bottom design: drain whole-squid liquid, hold paste	Wolfking	With preliminary GA
Frame base material (stainless or hot-dip galvanized) and price difference	Wolfking	With proposal
One hydraulic unit per freezer or one shared	Wolfking	With preliminary GA
Engine room: containerized vs. open skid, location, noise treatment	Del Mar + APCCO site meeting	Early November
Ammonia charge and CalARP position	Wolfking calculates, APCCO reviews	With refrigeration concept
Condenser type and water availability	Wolfking proposes, APCCO confirms	With refrigeration concept
Hot-gas defrost requirements and valve scope split	Wolfking + APCCO	With P&ID
Electrical load list for the PG&E application	Wolfking	As soon as possible
Phase-1 loading method	Del Mar decides, Wolfking advises	With GA
Shipping terms and port	Del Mar + Wolfking	With proposal
Installation duration and crew size	APCCO	After preliminary design

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